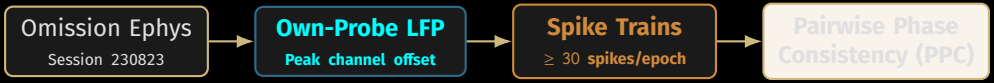
jnwb: Spike and Field

Spike-field phase coupling by area and epoch, from one omission session

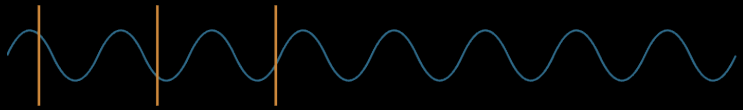
Hamed Nejat
Bastos Lab · Vanderbilt University

October 29, 2026



Does a spike fall at a particular phase of the local field?

Each orange tick is a spike. Phase consistency asks whether spikes cluster at one phase of the cycle.



Schematic Motif
Not empirical session data

Pairwise Phase Consistency (PPC)
Vinck et al. (2010), unbiased rhythmic metric

Three measures, one field per unit

PPC
Pairwise Phase Consistency
Unbiased by spike count (Vinck 2010). Primary measure here.

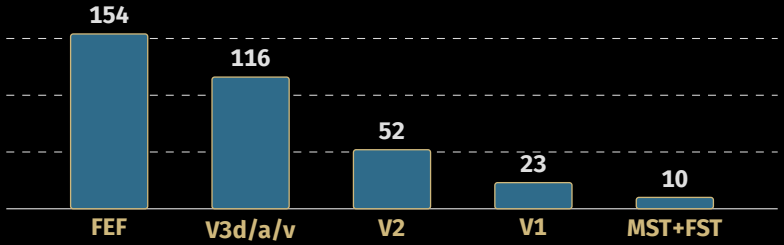
PLV
Phase-Locking Value
Derived as median of $\sqrt{\text{PPC}}$. Approximate proxy.

r_{cl}
Circular-Linear Correlation
Computed over whole task window, not partitioned per epoch.

Field Definition: Each unit's own-probe LFP, at a fixed offset from its peak channel, in its own area.
Inclusion Filter: Units require $\geq$ 30 spikes within an epoch. Sourced from Vinck et al. (2010, 2011).

Most units are in V3d/a/v and FEF

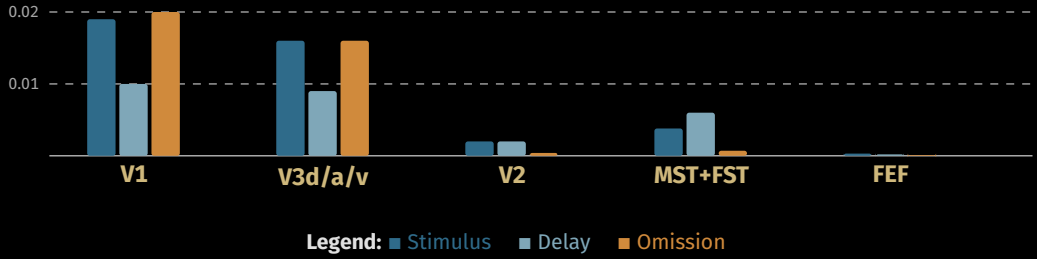
Units per area after 30-spike filter, stimulus epoch. Session `sub-C31o_ses-230823`.



368 units in bank; 5 without area label dropped. Units below 30 spikes in epoch excluded.

Theta locking is strongest in V1 and V3d/a/v, and holds in omission

Median PPC, theta 4–8 Hz across task epochs.



V1: 0.019 / 0.010 / 0.020 V3d/a/v: 0.016 / 0.009 / 0.016 FEF: 0.000 in every epoch.

Low gamma peaks in V3d/a/v; high gamma is weak except in V1 and V3

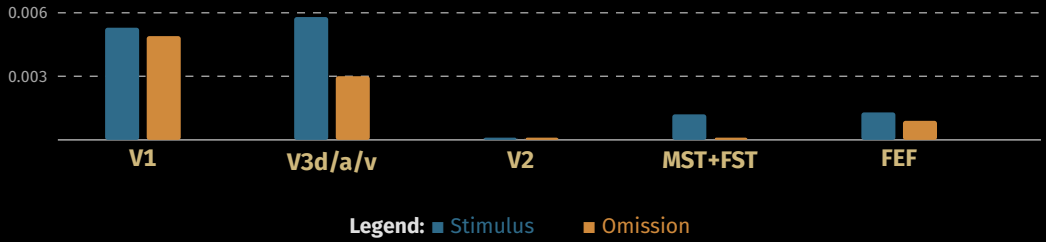
Median PPC, stimulus epoch. Heatmap shaded by magnitude.

Area	Theta (4–8 Hz)	Low γ (30–50 Hz)	High γ (50–80 Hz)
V1	**0.0192**	0.0020	0.0053
V3d/a/v	**0.0159**	**0.0269**	0.0058
V2	0.0020	0.0002	0.0000
MST+FST	0.0038	0.0017	0.0012
FEF	0.0003	0.0010	0.0013

Scale: $<$ 0.002 pale · 0.002–0.006 light · 0.006–0.015 mid · $\geq$ 0.015 dark (high locking).

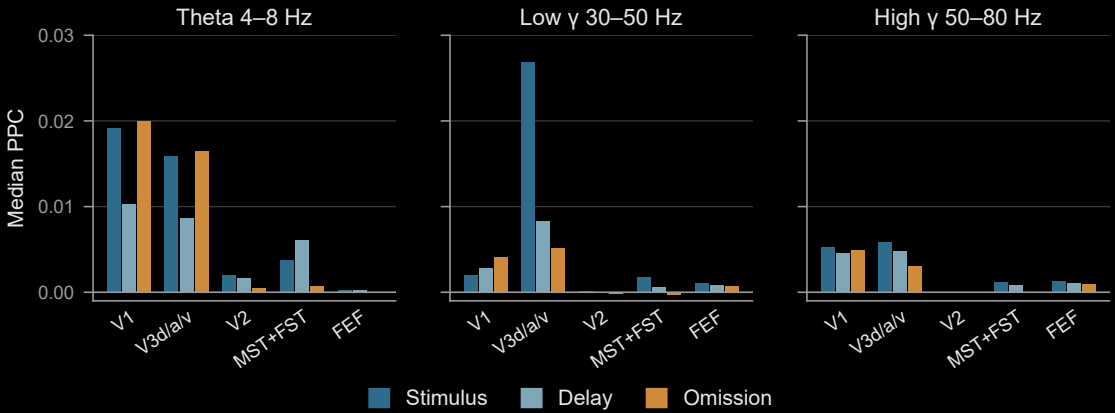
High gamma falls in omission in V3d/a/v, and holds in V1

Median PPC, high gamma 50–80 Hz. Stimulus vs. Omission epoch.



V3d/a/v falls from 0.006 to 0.003. V1 holds at 0.005. Omission epoch has fewer spikes per unit.

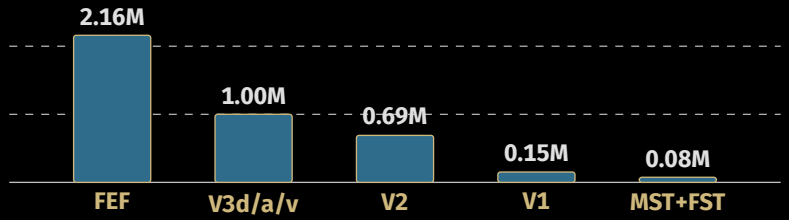
V1 and V3d/a/v lead in all nine band × epoch cells



Session sub-C31o_ses-230823 · units with ≥30 spikes in the epoch · own-probe LFP · omission unit feature bank

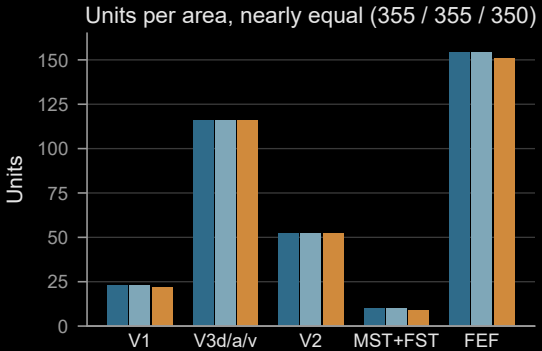
V1 and MST+FST rest on far fewer spikes

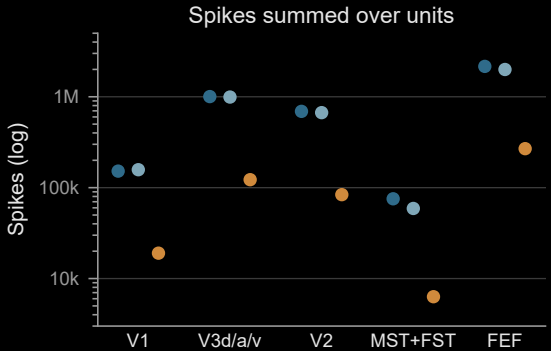
Spikes summed over units, stimulus epoch.



V1: 23 units with 152k spikes · MST+FST: 10 units with 75k spikes. Values are tentative.

Omission has 8× fewer spikes than stimulus





Stimulus Delay Omission

Session sub-C31o_ses-230823 · units with ≥30 spikes in the epoch · own-probe LFP · omission unit feature bank

Where the results are weak

V2 Area (52 units)
Theta PPC: 0.002 (stim/delay), 0.000 in omission.
Low/high γ near zero.

MST+FST (10 units)
Theta PPC: 0.004–0.006.
Sample size too restricted to resolve robust pattern.

FEF (154 units)
Near zero in all bands and epochs.
Largest unit bank; unambiguous session null.

A null in FEF is not a claim about FEF. It holds strictly for this field, this filter, and this session.
Field for every area: own-probe LFP at fixed offset from peak channel.

What these numbers cannot yet say

Single Session: Descriptive only; no multi-session hypothesis tests.

Bands: 4–8, 30–50, 50–80 Hz (not 4–12 or 30–80 Hz).

Derived PLV: Median of $\sqrt{\text{PPC}}$, not phase-computed.

Local Probes: No cross-area pairs; own-area LFP only.

Bank Provenance: Dated 2026-10-07; receipt flagged `producer-changed`. Values not yet regenerated.

Source: omission project, unit feature bank.

Decisions for the next run

1. Analytical Run

Per-epoch empirical PLV and cross-area spike-field coupling pairs.

2. Frequency Boundaries

Keep canonical project bands vs. align with 4–12 Hz and 30–80 Hz.

3. Replicate Session

Analyze second session to verify if V1/V3 theta locking replicates.

4. Regenerate Bank

Recompute unit feature bank to bring provenance receipt current.

Each decision is Hamm's. The omission project runs the analysis once decided.